

Neurodynamics-Driven Coupled Neural P Systems for Multi-Focus Image Fusion

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Abstract

Multi-focus image fusion (MFIF) is a crucial technique in image processing, with a key challenge being the generation of decision maps with precise boundaries. However, traditional methods based on heuristic rules and deep learning methods with black-box mechanisms are difficult to generate high-quality decision maps. To overcome this challenge, we introduce neurodynamics-driven coupled neural P (CNP) systems, which are third-generation neural computation models inspired by spiking mechanisms, to enhance the accuracy of decision maps. Specifically, we first conduct an in-depth analysis of the model's neurodynamics to identify the constraints between the network parameters and the input signals. This solid analysis avoids abnormal continuous firing of neurons and ensures the model accurately distinguishes between focused and unfocused regions, generating high-quality decision maps for MFIF. Based on this analysis, we propose a **Neurodynamics-Driven CNP Fusion model (ND-CNPFuse)** tailored for the challenging MFIF task. Unlike current ideas of decision map generation, ND-CNPFuse distinguishes between focused and unfocused regions by mapping the source image into interpretable spike matrices. By comparing the number of spikes, an accurate decision map can be generated directly without any post-processing. Extensive experimental results show that ND-CNPFuse achieves new state-of-the-art performance on four classical MFIF datasets, including Lytro, MFFW, MFI-WHU, and Real-MFF. The code is available at <https://github.com/MorvanLi/ND-CNPFuse>.

1. Introduction

Multi-focus image fusion (MFIF) [39] is an important task in computer vision that aims to generate an all-in-focus image by fusing multiple images of the same scene captured at varying focal depths. The fused image improves both perceptual quality and information richness, thereby providing

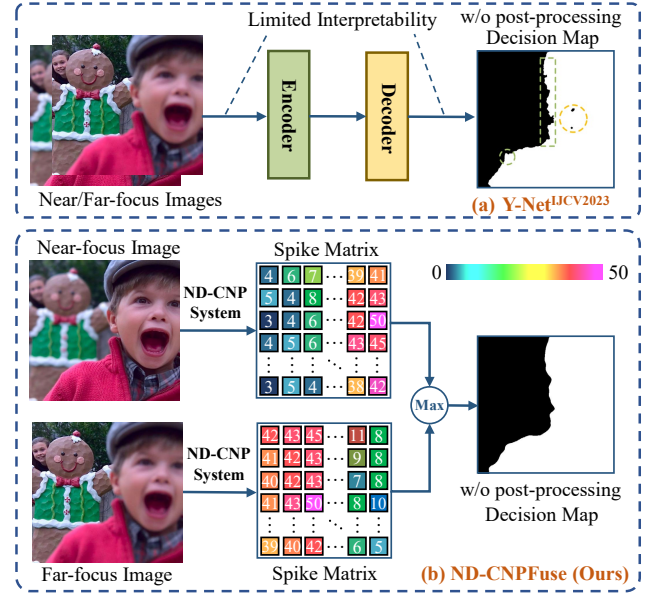


Figure 1. Workflow comparison between the existing DL-based decision method and the proposed ND-CNPFuse. ND-CNPFuse encodes source images into spike matrices and generates the decision map by comparing spike counts. Focused regions generate more spikes than unfocused ones, consistent with human visual perception [9].

a reliable input for a range of downstream vision applications such as action detection [27] and object detection [13]. By integrating the focused regions from source images with different focus settings, MFIF effectively alleviates the blurring artifacts introduced by the limited depth of field inherent in imaging systems.

Before the advent of deep learning (DL), MFIF methods were performed in the transform [14] and spatial domains [40]. The effectiveness of these methods often relies heavily on manually designed fusion rules and strong prior assumptions, which may not always be applicable or sufficiently accurate.

In recent years, DL-based approaches have become the dominant paradigm in MFIF, achieving remarkable

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progress [16]. Some studies have applied DL techniques to end-to-end image fusion frameworks [32, 34]. Different end-to-end methods may use various network structures, but they generally consist of an encoder network for extracting image features, a fusion network for merging features, and a decoder network for reconstructing the fused features. Another line of studies draws inspiration from spatial domain methods, suggesting the use of DL techniques to generate decision maps [19, 24, 28, 42]. This process involves training a network to map the source image to a focus map, ultimately producing decision maps. Compared to end-to-end methods, decision-based methods directly retain information from the source image at the pixel level, avoiding the spatial consistency issues of end-to-end methods. Therefore, more research has been done on DL-based decision map methods. However, DL-based decision map methods still have evident drawbacks. The most crucial one is that the internal working mechanism is difficult to explain, hindering a clear understanding of the mapping between source images and the decision map [29]. This will result in nuisance false edges and burrs in the generated decision map, as can be intuitively observed in Fig. 1(a).

Coupled neural P (CNP) systems [22] are a class of neural-like computing models inspired by the mechanism of spiking neurons, and are classified as third-generation neural networks. A CNP system generates synchronous spikes through the mutual coupling of neurons, where synchronous pulses of neurons are treated as 0 or 1. After multiple iterations, the system produces a sequence of binary images that record the firing states of each neuron at every time step. By comparing the spike counts of neurons, the degree of focus corresponding to each pixel can be effectively estimated, enabling the differentiation between focused and defocused regions. *This is particularly well-suited for decision map generation in the MFIF task, which serves as the core motivation for introducing CNP systems into MFIF.* However, when CNP systems are directly applied to MFIF, neurons may suffer abnormal continuous firing. In this case, the spike counts cannot accurately reflect the focus differences of the source images, which affects the quality of the decision map. This issue stems from an inappropriate constraint between network parameters and input signals.

To address this problem, we first conduct a detailed study on the neurodynamics of CNP neurons. This analysis reveals the underlying constraints between network parameters and input signals, thereby preventing abnormal continuous firing in neurons. A solid understanding of the neural firing mechanism is helpful for setting appropriate parameters and provides a theoretical basis for effectively applying CNP systems in MFIF. Building on the neurodynamics-driven CNP system, we then propose a new fusion method called **ND-CNPFuse**. As illustrated in Fig. 1(b), ND-CNPFuse provides an interpretable and accu-

rate decision map generation workflow, in contrast to existing DL-based MFIF methods that rely on black-box mechanisms (Fig. 1(a)).

In conclusion, this paper makes three contributions:

1. We conduct a detailed analysis of the neurodynamic mechanisms of CNP neurons, which is essential to clarify the constraints between network parameters and input signals. To the best of our knowledge, this is the first study to investigate the neurodynamics of CNP systems.
2. We propose a bio-inspired neural network-based method called ND-CNPFuse to address the challenge of generating accurate decision maps in MFIF tasks. The advantages of ND-CNPFuse are that it requires no training and the process of generating decision maps is interpretable.
3. Extensive experimental results show that ND-CNPFuse achieves state-of-the-art performance on four MFIF datasets. Furthermore, ND-CNPFuse consistently outperforms baseline CNP in six commonly used quantitative metrics, with an average improvement of 5.73%.

2. Related Work

2.1. DL-based MFIF methods

End-to-end methods. These methods utilize the powerful fitting capabilities of deep neural networks to directly generate fused images by learning features from the source images. Representative approaches include those based on convolutional neural networks [3, 31, 34, 35, 41], generative adversarial networks [6, 36], and Transformer architectures [20, 26, 43]. It is challenging for these methods to maintain the spatial consistency of the source images.

Decision-map methods. These methods formulate MFIF as a pixel-level classification task by training a deep neural network to predict decision maps for the focused and defocused regions. Liu et al. [15] pioneered the application of this technique in the MFIF task, but the accuracy of the generated decision maps is still limited. To improve the accuracy of decision maps, subsequent studies have introduced ensemble learning [1], attention mechanisms [30], regression pair learning [12], multi-scale architectures [17], edge preservation techniques [29], and explicit defocus blur modelling [25]. Due to the unclear working mechanism of the network, there are interfering pseudo-edges and burrs in the decision maps. How to generate high-quality decision maps for the MFIF task remains to be explored.

2.2. Coupled neural P systems

Coupled neural P (CNP) systems are a class of multi-parameter, single-layer, locally connected neural networks. Their key feature, the coupled spiking mechanism, enables efficient image feature extraction and has been applied in image fusion tasks [4, 10, 23]. However, existing CNP systems heavily rely on manually tuned parameters. This may

lead to neurons in invalid states, thus limiting the model's performance in image fusion tasks. In this paper, we analyze the dynamics of CNP neurons and establish the constraint relationships between network parameters and input signals, which can make the CNP system easier to use.

3. Neurodynamics-Driven CNP systems

In this section, we first review the definition of the CNP system and then analyze its neurodynamic properties in detail.

3.1. Mathematical model of CNP systems

In CNP systems, each neuron has three memory units: the feeding input unit (U), the linking input unit (V), and the dynamic threshold unit (T). The update rule for CNP neuron σ_{ij} states is defined as follows:

$$U_{ij}(t+1) = \begin{cases} U_{ij}(t) - u + I_{ij}(t) + \sum_{\sigma_{kl} \in \delta_r} W_{kl} P_{kl}(t), & \text{if } \sigma_{ij} \text{ fires} \\ U_{ij}(t) + I_{ij}(t) + \sum_{\sigma_{kl} \in \delta_r} W_{kl} P_{kl}(t), & \text{otherwise} \end{cases} \quad (1)$$

$$V_{ij}(t+1) = \begin{cases} V_{ij}(t) - v + \sum_{\sigma_{kl} \in \delta_r} W_{kl} P_{kl}(t), & \text{if } \sigma_{ij} \text{ fires} \\ V_{ij}(t) + \sum_{\sigma_{kl} \in \delta_r} W_{kl} P_{kl}(t), & \text{otherwise} \end{cases} \quad (2)$$

$$T_{ij}(t+1) = \begin{cases} T_{ij}(t) - \tau + \lambda P_{ij}(t), & \text{if } \sigma_{ij} \text{ fires} \\ T_{ij}(t), & \text{otherwise} \end{cases} \quad (3)$$

where subscripts (i, j) denote the coordinates of the neuron, (k, l) denote the coordinates of its neighboring neurons, and t represents the current iteration count. u, v, τ indicate the spikes consumed by three units. $I_{ij}(t)$ is an external input. W_{kl} characterizes synaptic weight. P_{kl} symbolizes spikes received from the surrounding neurons σ_{kl} , where neighborhood radius is r ($\sigma_{kl} \in \delta_r$). λ is a threshold weight.

3.2. Why neurodynamics analysis is necessary?

In certain cases, CNP neurons may enter a state of **continuous firing**, generating spikes at each time step. As illustrated in Fig. 2(c) and Fig. 2(d). This uncontrolled spiking behavior can lead to model failure, as the output no longer reflects the features of the input signal. For the MFIF task, such outputs do not effectively distinguish between focused and defocused regions, thereby affecting the generation of accurate decision maps. In the following, we analyze this invalid state and identify parameter configurations that can effectively prevent it.

Analysis of continuous firing condition. We analyze the continuous firing condition of CNP neurons to derive constraints between parameters and the input signal. Suppose t tends to infinity, all neurons fire. In this case, the state

update equation becomes:

$$U(t+1) = U(t) - u + I(t) + \sum_{\sigma_{kl} \in \delta_r} W_{kl} P_{kl}(t), \quad (4)$$

$$V(t+1) = V(t) - v + \sum_{\sigma_{kl} \in \delta_r} W_{kl} P_{kl}(t), \quad (5)$$

$$T(t+1) = T(t) - \tau + \lambda P(t). \quad (6)$$

According to the theory in [22], the spike consumption of the three units in CNP neurons exhibits a linear relationship. Thus, the consumption rules can be described as:

$$U(t) - u = \alpha U(t), V(t) - v = \beta V(t), T(t) - \tau = \gamma T(t), \quad (7)$$

where $\alpha, \beta, \gamma \in (0, 1)$. We have the following theorem:

Theorem 1. *For the feeding input unit U , the value increases with iterations and is represented as*

$$U(t) = I \frac{1 - \alpha^t}{1 - \alpha} + \sum_{i=0}^{t-1} K(i) \alpha^{t-i-1}. \quad (8)$$

Proof. Let us clarify the initial conditions: $U(0)$ and $K(0)$ are both zero. Define $K(i) = \sum_{\sigma_{kl} \in \delta_r} W_{kl} P_{kl}(t-1)$. Next, we describe $U(1), U(2), U(3), \dots, U(t)$ step by step:

$$\begin{aligned} U(0) &= 0 \\ U(1) &= \alpha U(0) + I + K(0) = I \\ U(2) &= \alpha U(1) + I + K(1) = \alpha I + I + K(1) \\ U(3) &= \alpha U(2) + I + K(2) = \alpha^2 I + \alpha I + I + \alpha K(1) + K(2) \\ &\vdots \end{aligned}$$

$$U(t) = I \frac{1 - \alpha^t}{1 - \alpha} + \sum_{i=0}^{t-1} K(i) \alpha^{t-i-1}. \quad (9)$$

Theorem 1 is proved. $\square$

Theorem 2. *For the linking input unit V , the value grows continuously with the increasing number of iterations and can be represented by the following formula:*

$$V(t) = \sum_{i=0}^{t-1} K(i) \beta^{t-i-1}. \quad (10)$$

Proof. Similarly, the initial state $V(0) = 0$. We list $V(1), V(2), V(3), V(4), \dots$, and $V(t)$.

$$\begin{aligned} V(0) &= 0 \\ V(1) &= \beta V(0) + K(0) = 0 \\ V(2) &= \beta V(1) + K(1) = K(1) \\ V(3) &= \beta V(2) + K(2) = \beta K(1) + K(2) \\ &\vdots \end{aligned} \quad (11)$$

$$V(t) = \sum_{i=0}^{t-1} K(i) \beta^{t-i-1}.$$

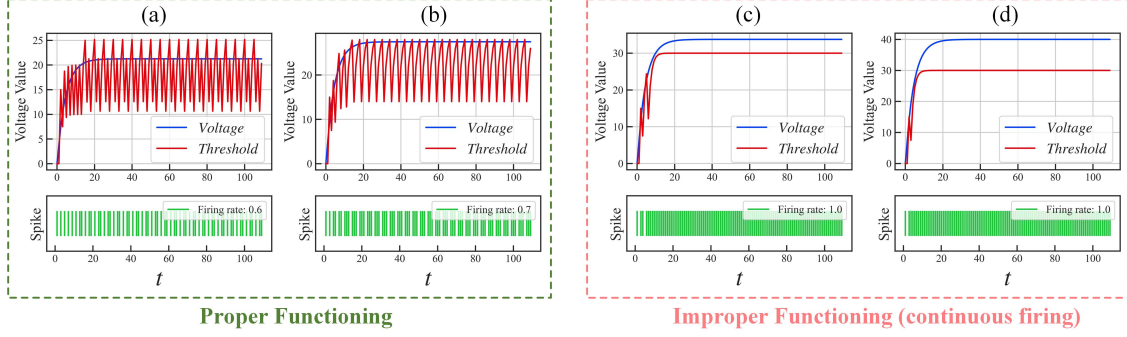


Figure 2. CNP neuron firing states on different inputs. (a) $I = 0.5$; (b) $I = 1.0$; (c) $I = 1.5$; and (d) $I = 2.0$.

Theorem 2 is proved. $\square$

Theorem 3. For the dynamic threshold unit T , the value increases with iterations and is expressed as follows

$$T(t) = \lambda \frac{1 - \gamma^{t-1}}{1 - \gamma}. \quad (12)$$

Proof. T denotes the dynamic threshold with initial state $T(0) = 0$, and P denotes the output with initial state $P(0) = 0$. A continuously firing neuron maintains $P(t) = 1$ as the firing condition always holds. Thus, we can get:

$$\begin{aligned} T(0) &= 0 \\ T(1) &= \gamma T(0) + \lambda P(0) = 0 \\ T(2) &= \gamma T(1) + \lambda P(1) = \lambda \\ T(3) &= \gamma T(2) + \lambda P(2) = \gamma\lambda + \lambda \\ &\vdots \\ T(t) &= \lambda \frac{1 - \gamma^{t-1}}{1 - \gamma}. \end{aligned} \quad (13)$$

Theorem 3 is proved. $\square$

Based on the above three basic theories, we can further obtain the condition that neurons are in sustained firing, i.e., the following Theorem 4.

Theorem 4. Relationship between external input and internal parameters of neurons in the continuous firing state

$$I > \frac{\lambda(1 - \alpha)(1 - \beta)}{(1 - \gamma)(1 - \beta + \text{sum}(W))} - \text{sum}(W). \quad (14)$$

Proof. Continuous firing requires the fundamental criterion $U(t) \times (1 + V(t)) > T(t)$ to hold at all times. Substituting results $U(t)$, $V(t)$ and $T(t)$ obtained by Theorems 1, 2 and 3, we can obtain:

$$\begin{aligned} \left(I \frac{1 - \alpha^t}{1 - \alpha} + \sum_{i=0}^{t-1} K(i) \alpha^{t-i-1} \right) \left(1 + \sum_{i=0}^{t-1} K(i) \beta^{t-i-1} \right) \\ > \lambda \frac{1 - \gamma^{t-1}}{1 - \gamma}. \end{aligned} \quad (15)$$

Given continuous firing, $K(i)$ is only related to the weights between neurons. At this point, $K(i)$ can be written as $\text{sum}(W)$ (constant synaptic input).

Eq. (15) can be expressed as:

$$\begin{aligned} \left(I \frac{1 - \alpha^t}{1 - \alpha} + \text{sum}(W) \frac{1 - \alpha^t}{1 - \alpha} \right) \left(1 + \text{sum}(W) \frac{1 - \beta^t}{1 - \beta} \right) \\ > \lambda \frac{1 - \gamma^{t-1}}{1 - \gamma}. \end{aligned} \quad (16)$$

When $t \rightarrow \infty$, we obtain Eq. (17)

$$I > \frac{\lambda(1 - \alpha)(1 - \beta)}{(1 - \gamma)(1 - \beta + \text{sum}(W))} - \text{sum}(W). \quad (17)$$

Theorem 4 is proved. More detailed derivations are provided in Appendix A.1. $\square$

To verify the accuracy of Theorem 4, we set up a group of values to test: $W = [0.1, 0.5, 0, 0.5, 0.1]$, $\alpha = 0.8$, $\beta = 0.2$, $\gamma = 0.5$, and $\lambda = 15$. The continuous firing condition is calculated based on Eq. (17).

$$I > \frac{15 \times 0.8 \times 0.2}{(1 - 0.5)(1 - 0.2 + 1.2)} - 1.2 = 1.2. \quad (18)$$

Fig. 2 shows the firing states of a CNP neuron for different input signals. When external input does not exceed I , the neuron can function properly, with firing rates of 60% and 70%, respectively. As shown in Fig. 2(a) and Fig. 2(b). In contrast, inputs exceeding I lead to continuous firing

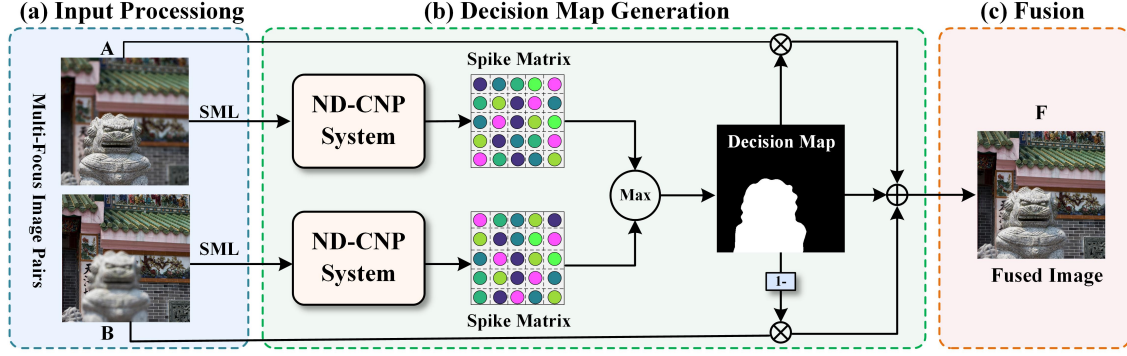


Figure 3. Overview of ND-CNPFuse for MFIF. Fused image F quality primarily relies on the decision map generated by the ND-CNP system. A and B are the multi-focus source images. $\otimes$ and $\oplus$ denote element-wise multiplication and addition.

with **100%** firing rates (Fig. 2(c) and Fig. 2(d)). The experimental results are consistent with our theoretical analysis, confirming the correctness of the mathematical derivation.

Corollary 1. *According to Theorem 4, the external input to a CNP neuron must remain below its internal state to ensure proper functionality, as demonstrated below*

$$I < \frac{\lambda(1-\alpha)(1-\beta)}{(1-\gamma)(1-\beta+\text{sum}(W))} - \text{sum}(W). \quad (19)$$

Corollary 1 reveals the necessary constraints between parameters and input signals for the proper functioning of CNP neurons. Detailed CNP parameter configurations (e.g., τ and λ) derived from Corollary 1 are provided in Appendix A.2. Noting that parameters are automatically configured by the system based on input MFIF images, eliminating the manual tuning and **demonstrating transferability and generalization across different datasets**. This helps to improve the ability of CNP systems to distinguish between focused and defocused regions in MFIF tasks, thereby improving the quality of the generated decision map.

4. Methodology

4.1. Problem statement

For a pair of multi-focus images A and B with the same size, MFIF aims to fuse the respective clear regions from images A and B to produce a fully focused image. An efficient and commonly used method is to generate a decision map (DM) of the source images. This pixel-level operation ensures that the fused image F maximizes the retention of information from the source images. As follows:

$$F(i, j) = A(i, j) \times \text{DM}(i, j) + B(i, j) \times (1 - \text{DM}(i, j)), \quad (20)$$

where (i, j) represents the pixel coordinates in the images.

Eq. (20) shows that DM accuracy directly impacts fused image quality. We employ neurodynamics-driven CNP

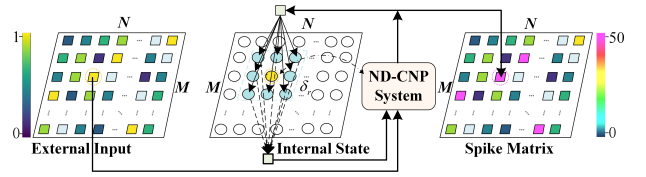


Figure 4. Working principle of the ND-CNP system. M and N denote image height and width. Please zoom in for a better view.

(ND-CNP) systems to generate accurate DM. Without loss of generality, we also provide details on handling **more than two input images** in Appendix B.

4.2. Overview of the architecture

The ND-CNPFuse framework is illustrated in Fig. 3, comprising three components: (a) input processing, (b) decision map generation based on the ND-CNP system, and (c) fusion. The details are described as follows.

(a) Input processing. The pixel values of the source image reflect only simple and low-dimensional information about the image, such as brightness or color. Using pixel values directly as input limits neuron firing. In contrast, image features can provide richer input signals. For this reason, we use sum-modified Laplacian (SML) [8] to preprocess the source image. Further analysis of the impact of SML on the entire framework is detailed in Sec. 5.4.

(b) Decision map generation. We use Φ_A and Φ_B to represent two ND-CNP systems associated with source multi-focus images A and B . The SML-preprocessed A and B serve as external inputs to Φ_A and Φ_B , respectively. Fig. 4 illustrates the detailed working principle of the model. The systems run from the initial state until reaching the maximum number of iterations. Subsequently, Φ_A and Φ_B output two spiking matrices, SM_A and SM_B , where each position in SM_A (or SM_B) represents the neuron firing counts in Φ_A (or Φ_B). Note that the mechanism of coupling neurons indicates that neurons exhibit cooperative firing phenomena (with a coupling radius of r) rather than indepen-



Figure 5. Qualitative results on “MFW-4” from MFW, with red and green boxes zoomed in 4 times for easy observation.

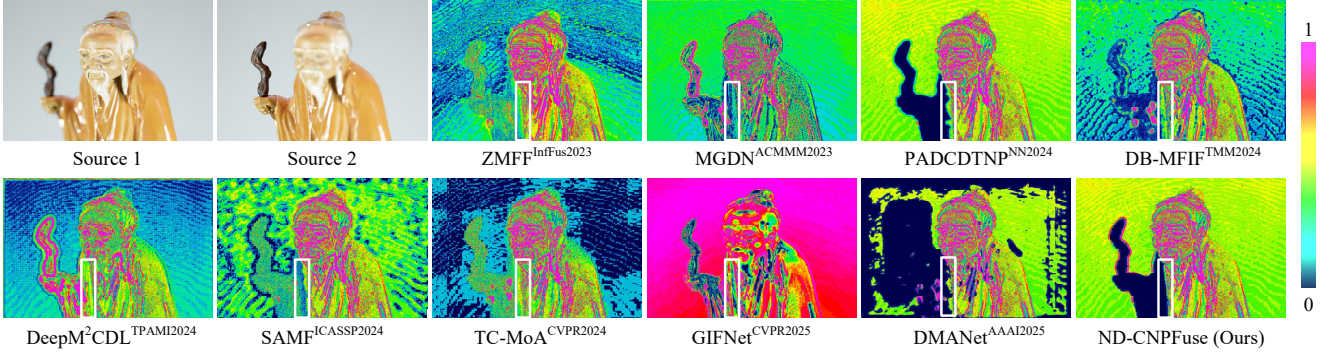


Figure 6. Visualization of difference images, obtained by subtracting Source 1 from the fused results. **Fewer residuals indicate better information preservation.** White boxes highlight regions for easy comparison. Zooming in for better observation.

dent firing. Therefore, based on SM_A and SM_B , the number of firing maps within their coupling radius r is denoted as F_A and F_B , respectively. As follows:

$$\begin{aligned} F_A(i, j) &= \sum_{(x, y) \in r(i, j)} SM_A(x, y), \\ F_B(i, j) &= \sum_{(x, y) \in r(i, j)} SM_B(x, y), \end{aligned} \quad (21)$$

where $r(i, j)$ is the coupling radius centered at position (i, j) . Further, we can obtain the DM:

$$DM(i, j) = \begin{cases} 1, & \text{if } F_A(i, j) > F_B(i, j) \\ 0, & \text{otherwise} \end{cases}. \quad (22)$$

(c) **Fusion.** After the ND-CNP system generates the DM, we can get the final fused image F according to Eq. (20).

5. Experiments

5.1. Experimental configurations

Dataset. To fully evaluate the effectiveness of NDCNPFuse, we conducted extensive experiments covering four datasets: Lytro [21], MFW [33], MFI-WHU [36], and

Real-MFF [37]. The Lytro dataset includes 20 pairs of color images. The MFI-WHU dataset consists of 120 pairs of artificially synthesized color images. The MFW dataset comprises 13 pairs of color images with defocus spread effects (DSE). The Real-MFF dataset contains 710 pairs of real-world MFIF images.

Comparison methods. We compare several SOTA methods on the MFIF task, including ZMFF [7], MGDN [5], PADCDTNP [11], DB-MFIF [38], DeepM²CDL [3], SAMF [13], TC-MoA [43], GIFNet [2], and DMANet [25]. These methods cover both representative end-to-end and decision-based approaches, providing a comprehensive baseline for evaluating fusion performance.

Evaluation metrics. We use six metrics to evaluate the fusion results from multiple perspectives: edge-based similarity measurement (Qabf), mutual information for wavelet feature (FMIw), structural similarity index (SSIM), peak signal-to-noise ratio (PSNR), phase congruency (QP), and human perception-inspired metric (QCB). For all these metrics, a higher value indicates better fusion quality [18].

Implementation details. ND-CNPFuse runs on an Intel(R) Core(TM) i5-13400 CPU, and DL methods run on an Nvidia A100 GPU. We set the coupling radius r to 16 and iterations t to 110. Please see Sec. 5.3 for detailed pa-

Table 1. Quantitative evaluation results of MFIF tasks. **Bold** and underlined indicate the highest and second-highest scores.

Multi-focus Image Fusion on Lytro							Multi-focus Image Fusion on MFFW						
	Qabf↑	FMIw↑	SSIM↑	PSNR↑	QP↑	QCB↑		Qabf↑	FMIw↑	SSIM↑	PSNR↑	QP↑	QCB↑
ZMFF	0.7031	0.3523	0.8395	26.7154	0.7853	0.7412	ZMFF	0.6655	0.3219	0.8148	23.6539	0.6469	0.6749
MGDN	0.7393	0.4558	<u>0.8505</u>	26.5215	0.8155	0.7020	MGDN	0.7013	0.4222	<u>0.8355</u>	23.7661	0.6773	0.6331
PADCDTNP	<u>0.7613</u>	<u>0.5916</u>	0.8417	26.9731	0.8394	<u>0.8075</u>	PADCDTNP	0.7367	<u>0.5268</u>	0.8224	23.1335	<u>0.7104</u>	0.7274
DB-MFIF	0.7479	0.5027	0.8402	26.5203	0.8398	0.7775	DB-MFIF	0.6958	0.4290	0.8207	23.3339	0.6620	0.6642
DeepM ² CDL	0.7181	0.4623	0.8387	26.0761	0.8092	0.7313	DeepM ² CDL	0.6718	0.4150	0.8311	23.3246	0.6664	0.6557
SAMF	0.7511	0.5585	0.8401	26.8899	0.8256	0.7951	SAMF	0.6292	0.3009	0.7984	23.2530	0.5586	0.6693
TC-MoA	0.7401	0.5274	0.8415	26.1720	0.8188	0.7597	TC-MoA	0.6034	0.2826	0.8216	23.4429	0.5307	0.6475
GIFNet	0.5194	0.3268	0.7854	25.9368	0.5412	0.6042	GIFNet	0.4509	0.2524	0.7653	22.5889	0.3850	0.5433
DMANet	0.7606	0.5807	0.8396	<u>26.9807</u>	<u>0.8439</u>	0.8055	DMANet	<u>0.7384</u>	0.5247	0.8119	23.6877	0.7073	<u>0.7279</u>
Ours	0.7621	0.5967	0.8541	26.9904	0.8466	0.8092	Ours	0.7399	0.5434	0.8362	<u>23.7154</u>	0.7294	0.7291

Multi-focus Image Fusion on MFI-WHU							Multi-focus Image Fusion on Real-MFF						
	Qabf↑	FMIw↑	SSIM↑	PSNR↑	QP↑	QCB↑		Qabf↑	FMIw↑	SSIM↑	PSNR↑	QP↑	QCB↑
ZMFF	0.6321	0.3811	0.8208	26.7501	0.6632	0.7093	ZMFF	0.7582	0.4081	0.9464	32.2025	0.8343	0.7567
MGDN	0.7230	0.5160	<u>0.8389</u>	27.1386	0.7527	0.7570	MGDN	0.8042	0.5050	<u>0.9552</u>	33.3076	0.8902	0.7849
PADCDTNP	0.7311	0.6193	0.8316	27.0261	0.7526	0.8231	PADCDTNP	0.8203	<u>0.6502</u>	0.9517	32.1256	<u>0.9213</u>	0.8318
DB-MFIF	0.7192	0.5278	0.8317	<u>27.6825</u>	0.7509	0.7813	DB-MFIF	0.7934	0.5353	0.9481	33.4508	0.8940	0.7990
DeepM ² CDL	0.6793	0.4992	0.8346	26.8603	0.7385	0.7426	DeepM ² CDL	0.7628	0.5124	0.9384	31.1958	0.8678	0.7784
SAMF	0.7263	<u>0.6248</u>	0.8304	27.6122	0.7571	0.8203	SAMF	0.7093	0.3293	0.9330	30.8761	0.7215	0.7707
TC-MoA	0.6557	0.5170	0.8388	27.6032	0.7098	0.7660	TC-MoA	0.7915	0.5109	0.9501	32.8831	0.8906	0.7971
GIFNet	0.4388	0.3536	0.7523	26.1136	0.4642	0.5500	GIFNet	0.4834	0.3572	0.8471	30.5346	0.6467	0.6041
DMANet	<u>0.7315</u>	<u>0.6248</u>	0.8302	27.6611	<u>0.7567</u>	<u>0.8237</u>	DMANet	<u>0.8205</u>	0.6484	0.9507	<u>34.0174</u>	0.9194	<u>0.8362</u>
Ours	0.7346	0.6268	0.8400	27.7140	0.7611	0.8265	Ours	0.8207	0.6666	0.9569	34.2024	0.9237	0.8398

Table 2. Runtime comparison of different fusion methods. Note that PADCDTNP, SAMF, and our proposed method are executed on an i5-13400 CPU, while the remaining methods are executed on an A100 GPU.

Method	ZMFF	MGDN	PADCDTNP	DB-MFIF	DeepM ² CDL	SAMF	TC-MoA	GIFNet	DMANet	Ours
Time (s)	74.48	<u>0.27</u>	1.35	0.47	14.35	0.43	1.24	0.56	0.21	0.41

parameter settings.

5.2. Performance comparison

Qualitative comparison. Fig. 5 shows the fusion results of different methods on the “MFFW-4” sample from the MFFW dataset. MGDN, PADCDTNP, DB-MFIF, TC-MoA, and our ND-CNPFuse effectively detect the boundary between focused and defocused regions, yielding good visual effects. However, the other methods encounter various issues. SAMF and DMANet suffer from edge diffusion artifacts. DeepM²CDL and GIFNet exhibit chromatic distortions. ZMFF loses some spatial details. The defocus spread effect present in the MFFW dataset poses additional challenges for producing high-quality fusion results. Nevertheless, ND-CNPFuse maintains more competitive fusion

performance compared to the other methods. *More qualitative fusion results* are provided in Appendix C.

We further visualize the difference images to compare fusion results, as shown in Fig. 6. Most methods show visible residuals, while PADCDTNP introduces boundary artifacts (see white boxes). In contrast, ND-CNPFuse fully preserves the focusing information from the source images, achieving more accurate boundaries.

Quantitative comparison. Tab. 1 reports the quantitative results of all methods on four datasets. ND-CNPFuse ranks first in five of six evaluation metrics across four datasets. It falls short of the best PSNR on MFFW as its spike count-based clarity measure may inadequately capture low-contrast edges, slightly impacting edge preservation. Nevertheless, it still ranks second. This indicates that

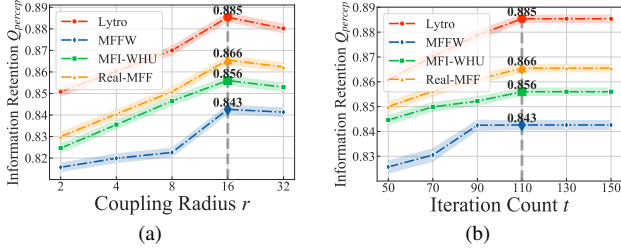


Figure 7. Effect of coupling radius and iteration count on the information retention metric Q_{percep} across four datasets.

Table 3. Ablation study results on the Lytro dataset. $\times$: without neurodynamic analysis, $\checkmark$: with neurodynamic analysis.

Config.	Qabf $\uparrow$	FMIw $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	QP $\uparrow$	QCB $\uparrow$
$\times$	0.747	0.509	0.841	25.702	0.835	0.754
$\checkmark$	0.762	0.597	0.854	26.990	0.847	0.809
Δ (%)	+2.02	+17.09	+1.56	+5.01	+1.45	+7.28

Table 4. Ablation study results on whether SML is used on Lytro.

Config.	Qabf $\uparrow$	FMIw $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	QP $\uparrow$	QCB $\uparrow$
$\times$ SML	0.761	0.593	0.852	26.987	0.848	0.808
$\checkmark$ SML	0.762	0.597	0.854	26.990	0.847	0.809

our method has good generalizability. We also acknowledge that the numerical improvement over the second-best method is not significant. However, the contribution of our work lies not only in performance gains but also in **offering a novel research perspective for the MFIF field**.

Runtime comparison. Tab. 2 lists the running times of all fusion methods on the Lytro dataset. Our method ranks third in runtime, but can achieve better performance within an acceptable execution time. Our ND-CNPFuse implementation is currently in MATLAB, and a re-implementation in C++ would further improve efficiency.

5.3. Parameter sensitivity analysis

We adopt the information retention Q_{percep} [18] to evaluate the effects of coupling radius and iteration count.

Impact of coupling radius. We test coupling radius values of 2, 4, 8, 16, and 32. As shown in Fig. 7(a), Q_{percep} increases with radius and peaks at 16, then declines due to loss of detail and gradient information. Thus, r is set to 16.

Impact of iteration count. Fig. 7(b) shows Q_{percep} results for t values of 50 to 150. Performance improves with more iterations and stabilizes around $t = 110$. To balance accuracy and efficiency, we set t to 110.

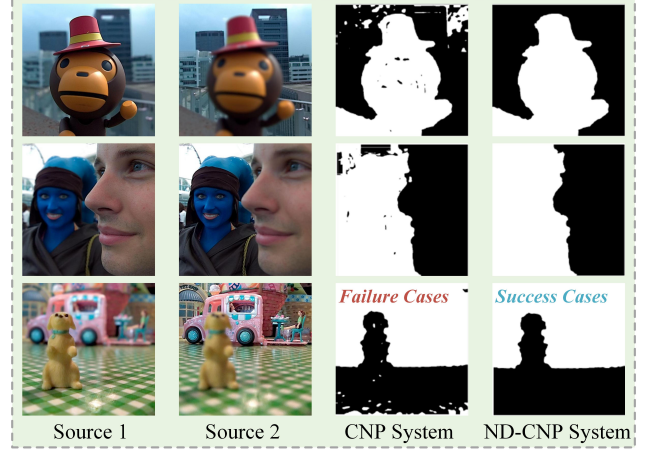


Figure 8. Visualization of decision maps for success cases (ND-CNP system) and failure cases (CNP system) on the Lytro dataset.

5.4. Ablation study

Neurodynamic analysis. We conducted ablation studies on the Lytro dataset to validate the effectiveness of neurodynamic analysis in improving model performance. Specifically, we compare the baseline CNP system and the neurodynamics-driven CNP system across six metrics, as summarized in Tab. 3. We can find that the neurodynamics-driven CNP system achieves significant improvements over the baseline across six multidimensional image quality metrics, with improvements of 2.02%, 17.09%, 1.56%, 5.01%, 1.45%, and 7.28%. These findings confirm the effectiveness of neurodynamic analysis.

SML module. We further validated the impact of the SML module on overall architecture performance. According to the quantitative analysis results in Tab. 4, after removing the SML module, evaluation metrics showed only a slight downward trend (e.g., PSNR decreased by 0.03 dB, and SSIM decreased by less than 0.02). This confirms the generalizability of the proposed method. Of course, by applying SML, the quality of fusion can be further enhanced.

We believe that ND-CNPFuse is unrestricted in the level of generalizability. First, neurons all parameters are automatically configured via neurodynamics. Second, the introduction or removal of the SML module has no significant effect on the overall framework’s performance.

5.5. Case study

We perform a qualitative case study on some samples from the Lytro dataset. Fig. 8 shows decision maps from the baseline and ND-CNP systems. The ND-CNP system produces decision maps with clearer boundaries and higher accuracy. In contrast, the original CNP system has uncontrolled spike firing in some areas, resulting in obvious misjudgments in the decision maps.

6. Conclusion

In this paper, we propose a novel MFIF method called ND-CNPFuse based on neurodynamics-driven CNP systems. The main contributions of ND-CNPFuse have two aspects. The first is that we provide a detailed theoretical neurodynamic analysis of the working mechanism of CNP neurons. This analysis reveals the constraints between network parameters and input signals, which helps prevent model failure caused by continuous firing. The second one is that we use spike counts of neurons to distinguish between focused and unfocused regions to generate decision maps without any post-processing, and the process is interpretable. Extensive experiments with nine SOTA fusion methods on four datasets show that our method is more competitive and generalizable.

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